

Real-time C+L band Transmission of 64 Tb/s with 1 Tb/s Single-carrier Channels over 1260 km

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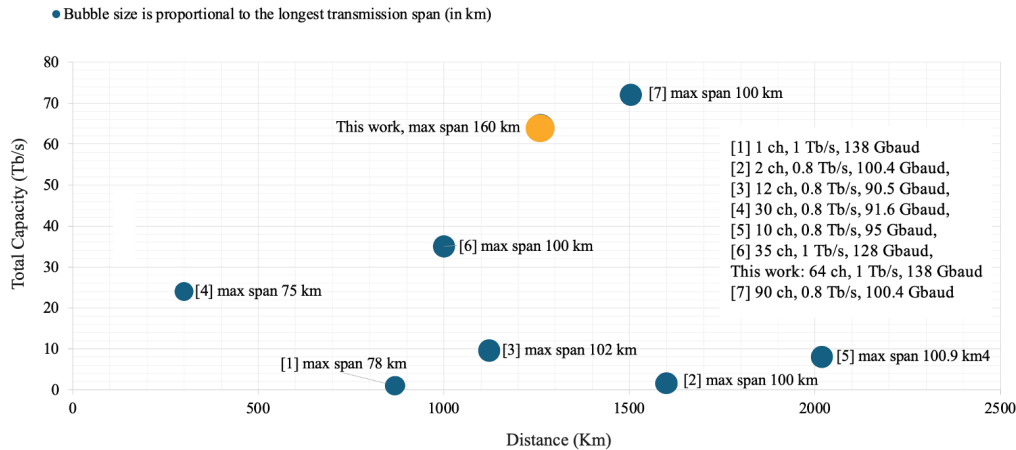
Abstract: We demonstrate real-time 64×1 Tb/s transmission over 1260km with variable ultra-long spans up to 160km. The 64Tb/s is enabled by ultra-low-loss, large effective area fiber, optimized hybrid Raman/EDFA amplification design combined with system level optimization.

1. Introduction

The evolution of optical communications continues to push the boundaries of transmission capacity and per-channel data rates, driven largely by advancements in Dense Wavelength Division Multiplexing (DWDM) [1]. As modern coherent technologies approach the Shannon limit, system-level factors, such as fiber type, amplification technologies, and interface technologies, have become increasingly critical for managing noise and maximizing throughput. These components collectively drive ongoing enhancements in DWDM performance. Figure 1 illustrates this progress by comparing real-time transmission experiments featuring per-channel data rates of 800 Gb/s or higher [1-7].

In this work, we demonstrate 64 Tb/s (1Tb/s x 64 channels) over 1260 km with variable and ultra-long span lengths of up to 160 km. The system employs commercial real-time 138 Gbaud Probabilistic Constellation Shaping 64 Quadrature Amplitude Modulation (PCS-64QAM) transceivers and commercial C- and L-band EDFA and hybrid Raman-EDFA line system, covering a total bandwidth of 9.6 THz. This achievement is enabled by a combination of ultra-low-loss, large-effective-area (A_{eff}) optical fiber, hybrid Raman/EDFA amplifier combined, and system level optimization, including per-band launch power and Raman pump configuration. Individual pump power optimization mitigates wavelength-dependent effects, such as, attenuation and nonlinearities, including the inter-channel stimulated Raman scattering (ISRS). This approach minimizes SNR variation across the full 9.6 THz bandwidth, enabling error-free 1 Tb/s transmission per channel over 1260 km. To the best of our knowledge, this represents the highest total fiber capacity achieved over a transmission distance exceeding 1000 km, using commercial real-time transponders and variable ultra-long span lengths of up to 160 km.

Fig. 1. Reported optical transmission experiments showing capacity versus distance achieved.



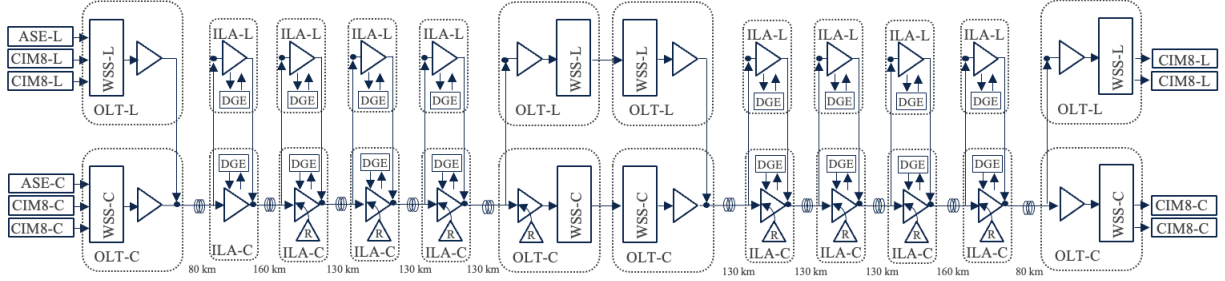
2. Transmission setup

Fig. 2 illustrates the experimental setup. At the transmitter site, two commercial real-time transponders per band (2.4T Line Cards equipped with Coherent Interconnect Module 8 - CIM8 optics) generate the data channels. Each channel is modulated with PCS-32QAM at a symbol rate of 138 Gbaud, yielding 4.4 bits/symbol and employing 15% soft decision-forward error correction (SD-FEC). The channels are spaced at 150 GHz, with each transponder configured

with a client rate of 400 Gigabit Ethernet (GbE) and a line rate of 1 Tb/s, enabling the transport of 5 x 400 GbE streams over 2 x 1 Tb/s channels.

To emulate full system loading, a built-in amplified spontaneous emission (ASE) loading mechanism generates C- and L-bands noise on a 75 GHz grid, covering a total bandwidth of 9.6 THz [8]. The ASE grid of 75 GHz is a system design choice, based on the default spectrum grid and segment typically allocated to accommodate 400 Gbps coherent optics carriers. The real-time channels are tuned across the C+L band, allowing the measurement of 64 x 138 Gbaud channels. Power levels are automatically adjusted via an optical channel monitor (OCM) to maintain a constant power spectral density (PSD). This channel replacement process is managed by the Wavelength Switching System (WSS) located on the head-end Optical Line Terminal (OLT) nodes.

Fig. 2. The schematic of the transmission testbed system.



The transmission line system uses two OLTs EDFA based C+L band units, two intermediate Raman and EDFA C+L band OLTs, and eight In-Line Amplifiers (ILA) using both Raman and EDFA technologies. To optimize lab resources, the testbed uses a loopback configuration: only half the hardware and fiber are deployed physically, with a loop at the OLT-C after 630 Km, effectively replicating the first half of the system (see Fig. 2).

Dynamic Gain Equalization (DGE) in ILAs maintains signal shape, while Automatic Power Control (APC) regulates Power Spectral Density (PSD) across the C+L bands on a span-by-span basis. The counter-propagating Raman modules uses five pumps at 1424 nm, 1438 nm, 1457 nm, 1470 nm, and 1495 nm to support C+L-Band operation. The transmission optical fiber is ITU-T G.654.E-compliant, with 0.150 dB/km average attenuation at 1550 nm and 125 μm^2 effective area. To emulate a real-field deployment, the link comprises eight spans of varying lengths (80 km to 160 km), totalling 1260 km. EDFA only amplification is used for 80 km spans, while hybrid Raman/EDFA is used for 130 km and 160 km span.

3. System design considerations and results

To ensure optimal transmission performance, the signal launch power was carefully designed per span and per band, with independent PSD targets set for each 12.5 GHz segment of the spectrum. This granular approach enabled per-band power optimization, helping to minimize SNR variation across the full C+L band and maintain consistent Quality of Transmission (QoT) across all wavelengths. The system design optimization included per band launch power and Raman pump power optimization to address wavelength-dependent attenuation and nonlinear effects, such as inter-channel stimulated Raman scattering.

Table 1(a) summarizes the average PSD per band (C and L), tilt across 4.8 THz, and total launch power for each span length (80 km, 130 km, and 160 km). The tilt is calculated as the power difference across the band from high to low frequencies. Table 1(b) details the design-target pump wavelengths, the corresponding fiber attenuation, and pump power values used for the 160 km span. The total power was 1.204 W with each pump operating below 400 mW.

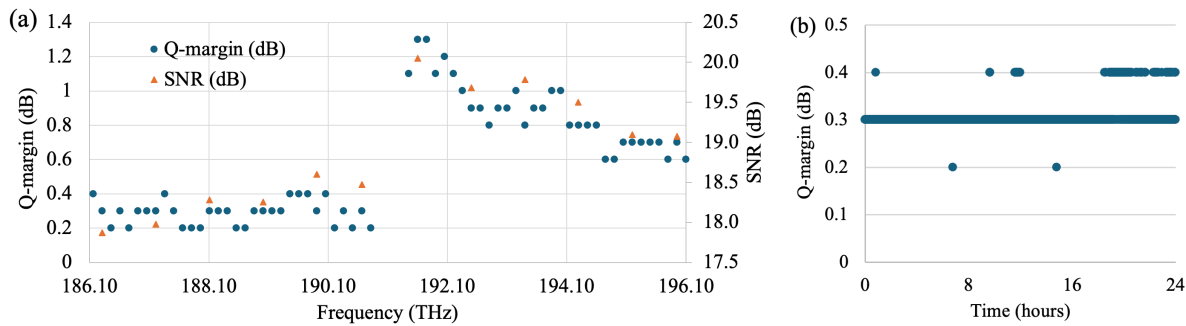
Table 1. (a) Average Power Spectral Density, Tilt, Total Launch Power. (b) Raman pump wavelengths, fiber attenuation coefficient, and Raman pump power values for the 160 km span length.

(a)	Span (km)	Avg. PSD/Band (dBm/12.5GHz)		Tilt/Band (dB/4.8THz)		Launch power Per band (dBm)		(b)	Wavelength (nm)	Fiber loss (dB/km)	Power (mW)
		C	L	C	L	C	L				
	80	-4.8	-6.3	-0.7	-0.4	19.5	18.5		1424	0.24	390
	130	-2.7	-4.6	-2.6	-0.9	21.7	20.9		1438	0.21	363
	160	-2.7	-4.6	-2.6	-0.9	21.7	20.9		1457	0.19	41
									1470	0.17	220

The average noise figure of the Hybrid Raman/EDFA amplification system was -1 dBm, with an average Raman gain per span of 10.8 dB.

Figure 3(a) shows the performance of each 1 Tb/s channel after 1260 km. The Q-factor margin (Q-margin) varied across a range of 1.1 dB, with minimum and maximum values of 0.2 dB and 1.3 dB, respectively. The Q-factor, is calculated as $Q\text{-factor (dB)} = 20 \times \log_{10} [-\text{norminv}(\text{BER})]$, where BER represents the pre-forward error correction bit error rate (pre-FEC BER), and “norminv” is the inverse cumulative distribution function (CDF) of the standard normal distribution. The SNR, measured over a 12.5 GHz integration bandwidth with 6 samples taken per band, ranged from 17.9 dB to 20 dB, with a total variation of 2.1 dB. Fig. 3(b) shows Q-margin measured over 24-hour period, for the worst-case channel at 190.81 THz, demonstrating transmission performance stability of 1 Tb/s per channel with total throughput of 64 Tb/s.

Fig. 3. (a) Per channel Q-margin and SNR recorded after 1260 km. The Q-margin 0.1 dB step size corresponds to the granularity offered by the platform's performance monitoring interface. (b) Stability measurement performed on worst-case channel over 24 hours.



4. Conclusion

Achieving homogeneous transmission performance across the C+L band is essential for practical deployment of long-haul terrestrial networks. This work demonstrates the feasibility of 150 GHz channel spacing for transporting 1 Tb/s per wavelength over 1260 km, with ultra-long span lengths of up to 160 km - a configuration that emulates real-world deployment scenarios with spans ranging from 80 km to 160 km.

This achievement was enabled by a combination of ultra-low-loss, large-effective-area fiber hybrid Raman/EDFA designing optimization, and system design optimization, including, per span and per-band launch power optimization and Raman amplifier design. Together, these elements enabled 1Tb/s error-free transmission across all 64 channels, achieving a total fiber capacity of 64 Tb/s over 1260 km.

5. References

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